ISRO

Indian Space Research Organisation

WEBSITE AUDIT REPORT

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1. Introduction

1.1 Introduction to Report

This report includes a UX audit of the ISRO website evaluating the key aspects like usability, accessibility, visual design, branding, performance, interaction, navigation, and a competitive analysis of the product ISRO. For this UX audit, the following pages will be evaluated: Home, About, Missions accomplished, one body content page of the website and also navigation

1.2 Introduction to Organisation

The Indian Space Research Organisation (ISRO) is an Indian space agency run by the government of India. The official website is https://www.isro.gov.in/. It is used to communicate major space mission reports, and research papers, and publish information. Also, publish opportunities for students and space enthusiasts. "web traffic has increased by 14.03% compared to last month. All traffic & engagement stats for isro.gov.in in the past 3 months below"(*Isro.gov.in Analytics,* 2025) with Total Visits 1.1M, Bounce Rate 68.36% Pages per Visit 2.79 Avg Visit Duration of only 00:01:27

1.3 UX Audit & Methodology

UX Audits are essential for checking website user experience and identifying the scope of enhancement of UX. Using heuristics principles and UX theories (Yablonski, 2020) like Hick's Law and Miller's Law makes decision-making simple and decreases cognitive load. Fitts's Law and Jakob's Law increase usability and provide an easy-to-navigate user interface. Gestalt Principles and the Law of Prägnanz facilitate well-structured and digestible content and design. Aesthetic-Usability Effect to improve the site, making it more aesthetic and up-to-date trends and providing a good experience to the user. This auditing will improve usability, responsiveness, and overall user experience, making the website efficient and usable. Using the design thinking process the UX audit is done to empathise and define the problem for the user (Moran & Brown, 2024)

2. Target Audience

The ISRO website has a diverse audience (*Isro.gov.in Analytics,* 2025)

1. **Scientists & Researchers** - Space scientists, engineers, and academics
2. **Students & Educators** - Students, career opportunities.
3. **Space Enthusiasts & General Public** – People who follow space missions and updates.
4. **Government & Industry Professionals** – Policymakers, businesses, and organisations.
5. **Media & Journalists** – Reporters covering ISRO missions & space exploration.

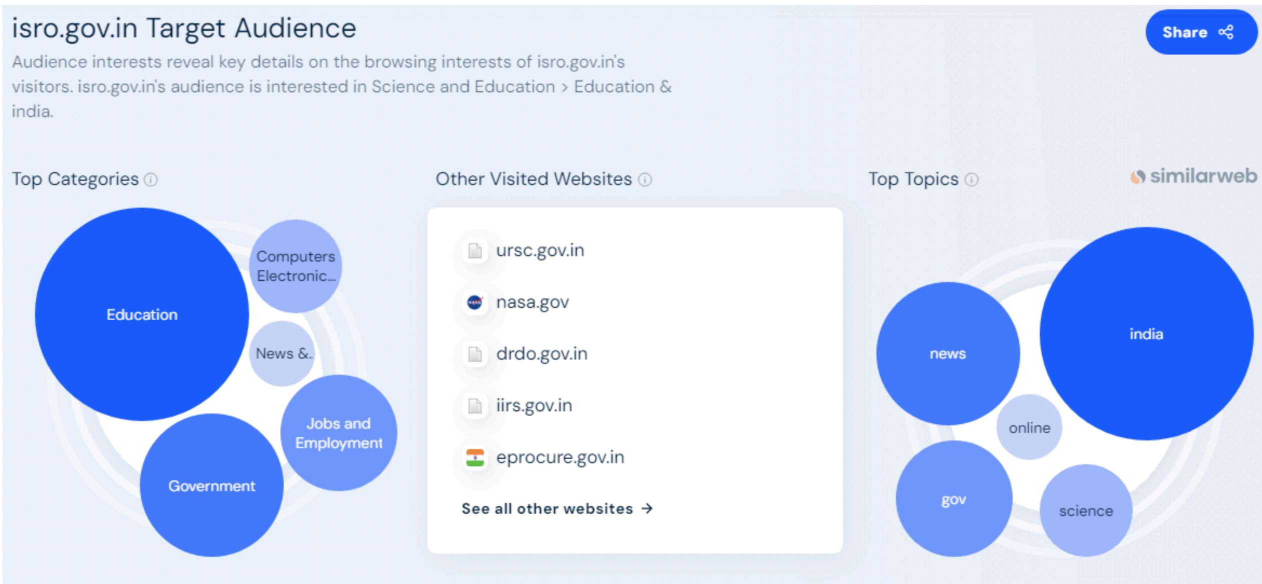


Fig. 1.1 (Target Audience analytics)(*Isro.gov.in Analytics,* 2025)

2.1 User Persona (Kaplan, 2022)

A user persona is a fictional representation of the target group (Kaplan, 2022) to understand the user needs like the goals and points.

Aarav Menon

AGE	25
EDUCATION	Masters
STATUS	Single
OCCUPATION	Aerospace Engineering Student
LOCATION	India
TECH LITERATE	High

Personality

Introvert · Thinker · Spender · Tech-savy

Bio

Aarav is a university student passionate about space exploration and dreams of working at ISRO. He follows every ISRO mission closely and frequently visits the website for updates, research papers, and internship opportunities.

Core needs

- Wants detailed mission updates and launch schedules.
- Seeks internship and career opportunities.
- Interested in educational resources and research papers.

Frustrations

- Finds it hard to locate internship applications and research papers.
- Wants a more interactive way to explore ISRO's missions.
- Not much choice and comparison not available

Brands

NASA · HUAWEI · facebook · YouTube · Instagram · ISRO · SPACEX · ROSCOSMOS

Fig. 1.2 (Persona of a student using ISRO website)

Hiroshi Tanaka

AGE	35
EDUCATION	Phd
STATUS	Married
OCCUPATION	Space Technology Researcher
LOCATION	Japan
TECH LITERATE	High

Personality

Introvert · Thinker · Spender · Tech-savy

Bio

Hiroshi is a researcher at JAXA (Japan Aerospace Exploration Agency), working on satellite technology. He follows ISRO's advancements in satellite launches, Chandrayaan, and Gaganyaan projects for potential collaboration.

Core needs

- Wants detailed technical specifications of ISRO's satellites and projects.
- Seeks collaboration opportunities between ISRO and JAXA.
- Interested in ISRO's research publications.

Frustrations

- Technical documents and data are not always publicly available.
- Collaboration details are not clearly listed on the website.
- Not much choice to get connected

Brands

NASA · HUAWEI · facebook · YouTube · Instagram · ISRO · JAXA · ROSCOSMOS

Fig. 1.3 (Persona of a researcher using ISRO website)

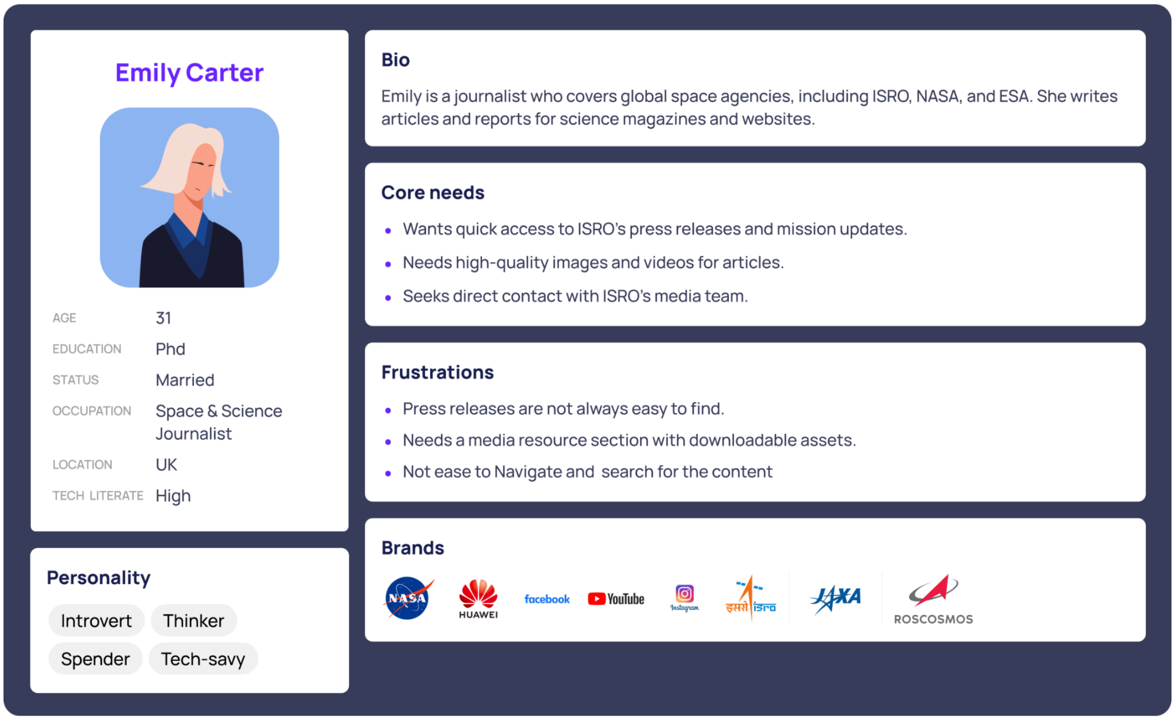


Fig. 1.4 (Persona of a journalist using ISRO website)

2.2 User journey map (Nielsen Norman Group, 2019)

The user journey maps are based on the student as the main target group searching for opportunities. The user journey map helps represent user experience and makes it easy to understand the frustration of the user (Nielsen Norman Group, 2019)

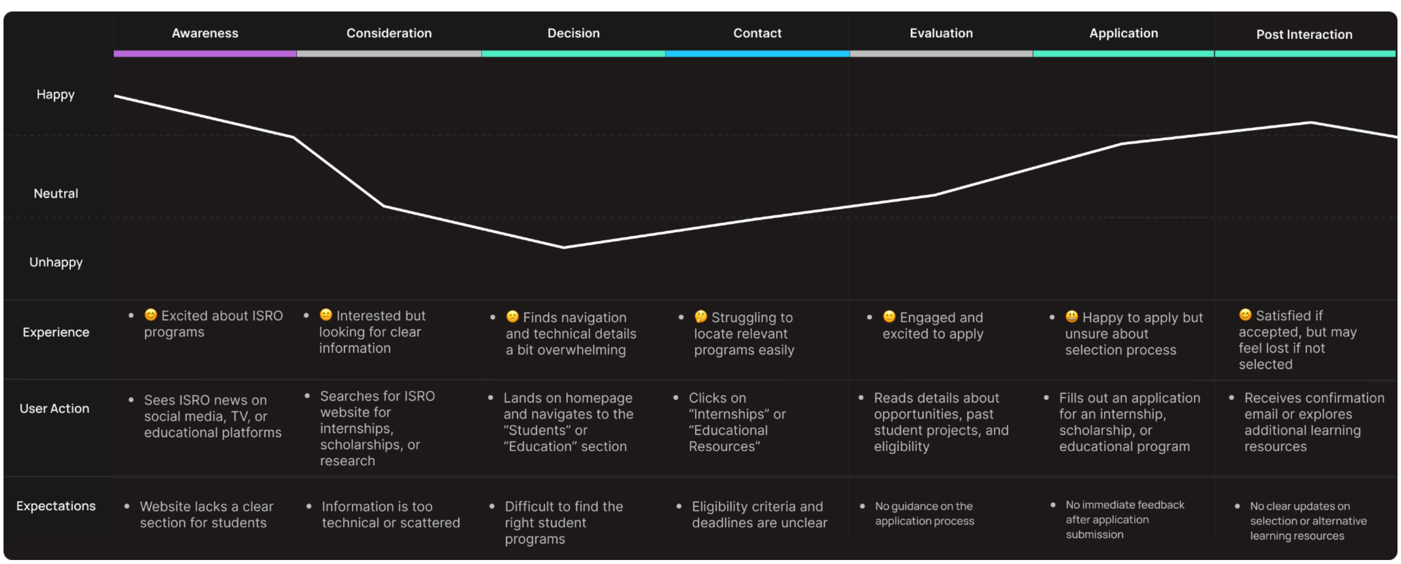


Fig. 1.5 (User journey map of a student exploring opportunities)

As the excited emotions decline as presented with unclear information difficult navigation and lack of guidance and they struggle to find a relevant program and information about application and eligibility the engagement improvement during application is uncertain both in the selection and feedback and post-interaction is like a relief after a task is done in a complicated website

Refer appendix for 2 more user journey maps

3. UX/UI Audit Analysis

3.1 Initial Impression (Desktop & Mobile)

To begin with, the first impression of the website is outdated UI from the 2000s The Home, About, Missions, and Body page non of pages are up to the trend and look outdated

3.1.1 Navigation & Information Architecture

In navigation, simplify interfaces to optimise the user experience. Excessive choices delay the process of decision-making while navigating a complicated system. According to Hick's Law: The time required to make a decision increases with the number and complexity of choices. (Yablonski, 2020)

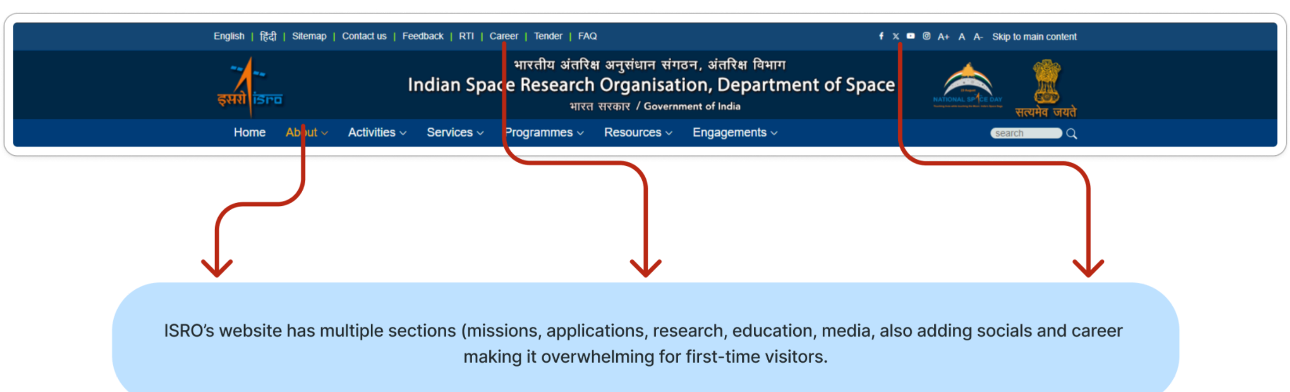


Fig. 1.6 Main Navigation Header

Cluttered Navigation: The header has 78 different pages to choose from (ISRO, 2019), making it hard to navigate with 6 main head sections, and each has at least 11 subsections to choose from. It creates frustration for the user

User Heuristics: Visibility of System Status & Error Prevention (Nielsen, 2024)

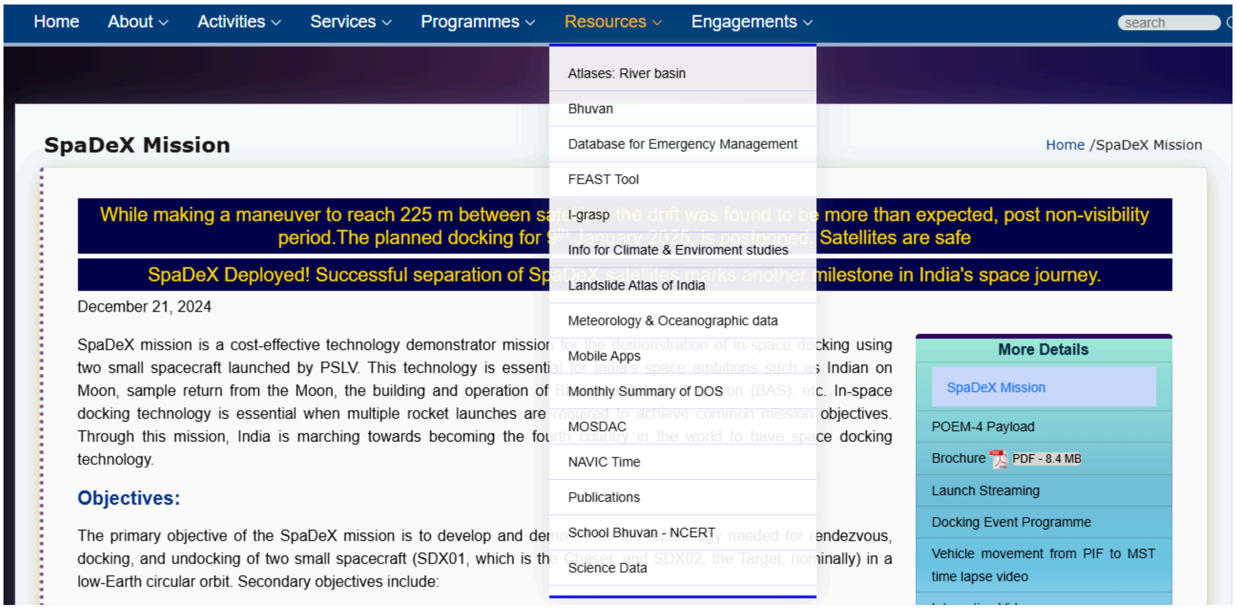


Fig. 1.7 Navigation subheading (Resources)

Recommendation: For ISRO's Navigation Header, this would mean designing intuitive, thin, and simple interfaces that minimise the cognitive load so that users can make decisions quickly, even under stress. Maintaining a consistent navigation structure across all pages ensuring clear labelling of links and buttons.

In the mobile menu to other portals, interactive items like call-to-action buttons and links need to be optimised. A small text-filled button that is far smaller than the user's thumb or finger can slow interaction, leading to frustration. According to Fitts's Law, the time to acquire a target is a function of its distance and size. (Yablonski, 2020)

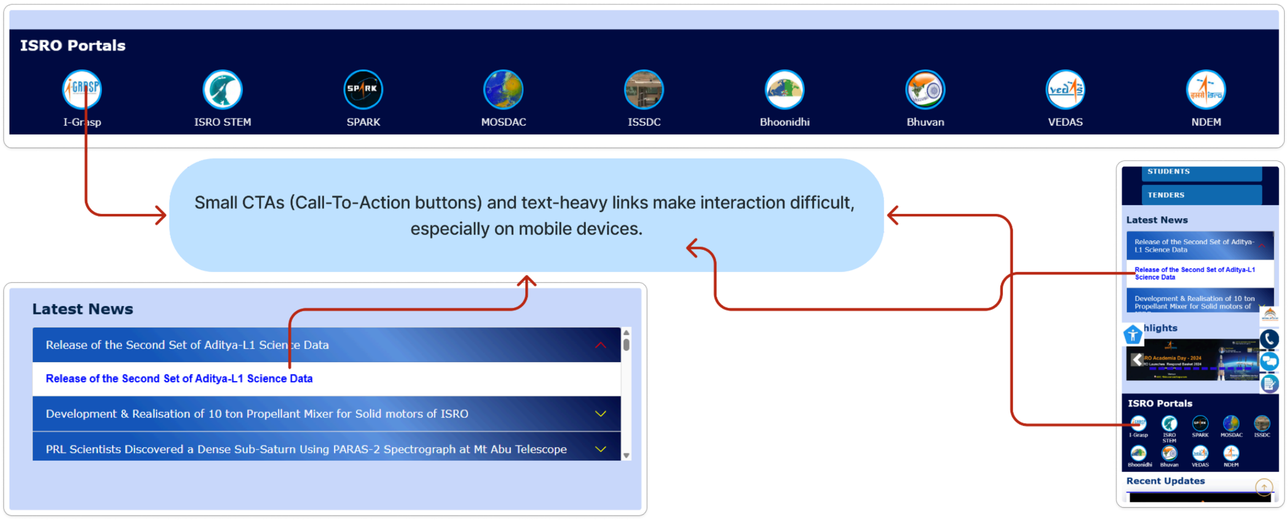


Fig. 1.8 ISRO Portal menu (Resources)

Recommendation: In terms of usability, it is significant to make more important and usable buttons that are within easy reach and have short and simple names, ensuring a fast and smooth experience on mobile platforms. and clear labelling of clickable links in text

3.1.2 Color, Branding & Visual

Although dashboard interfaces are functionally capable, the old visual style from the 2000s and the disorienting and with chaotic colour palette gives an impression of inefficiency. According to the Aesthetic-Usability Effect, Users perceive aesthetically pleasing designs as more usable, even if they are not. (Stevens, 2024)

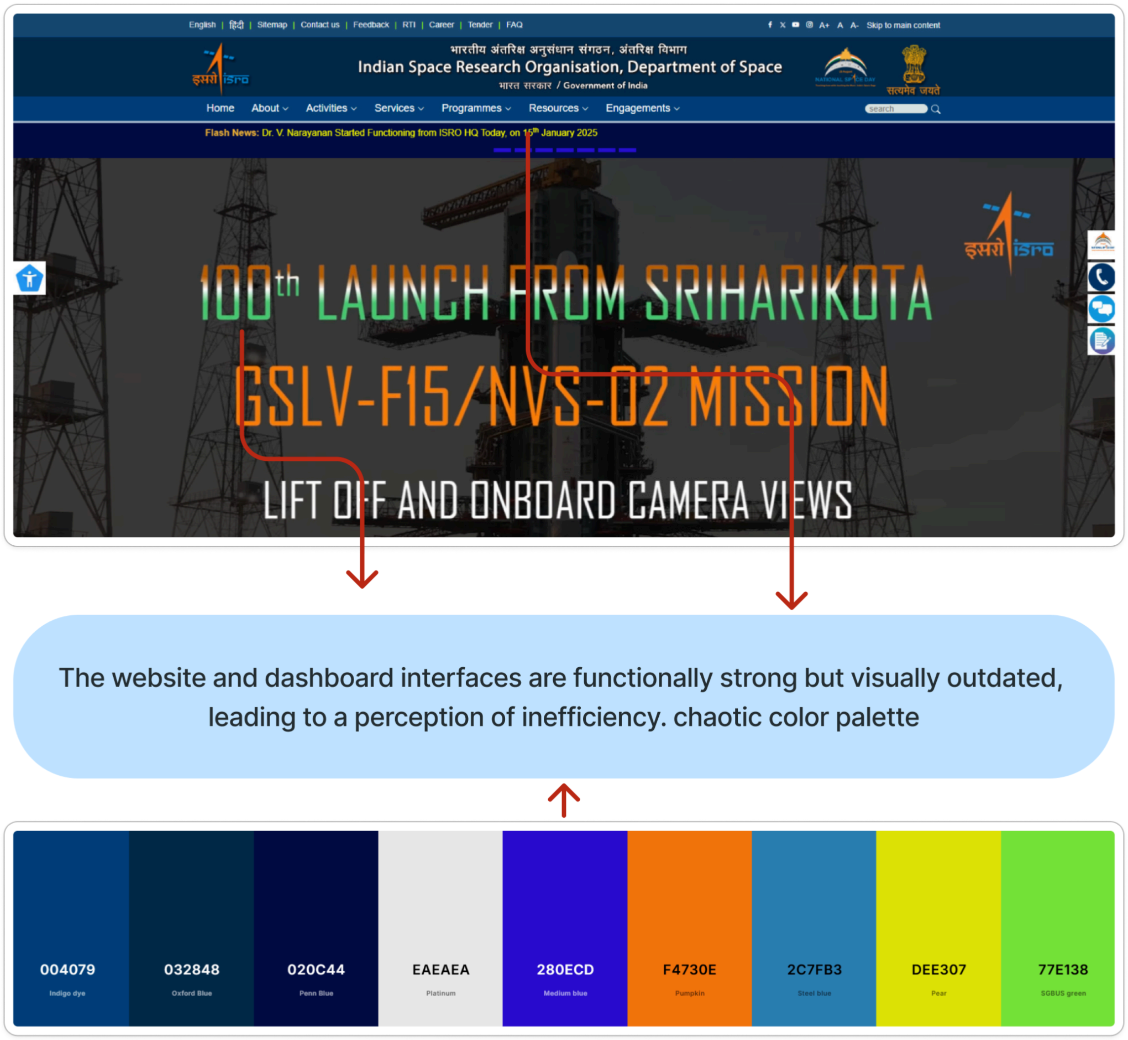


Fig. 1.9 Home page with the colour palette

User Heuristics: Match between the system and the real world & Consistency and Standards (Nielsen, 2024)

Recommendation: For improved user experience, a redesign of the visual style with a more harmonious, modern style would not just make the site more appealing but also improve usability, leading to improved overall user experience.

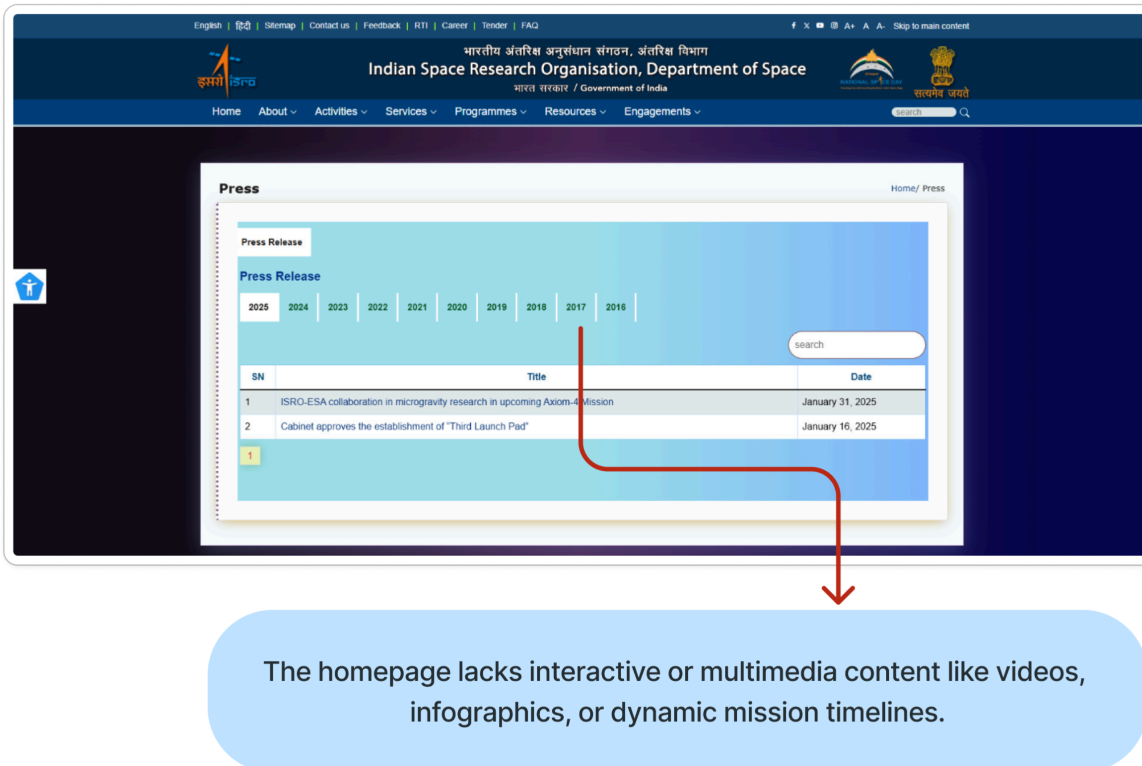


Fig. 2.1 Home page with press release timeline

The interface might be less intuitive or harder to use for the user, yet it is operational, and the homepage lacks interactive or multimedia content like videos, infographics, or dynamic mission timelines. There is no consistency in iconography and no brand language throughout the website

User Heuristics: Consistency and Standards: (Nielsen, 2024)

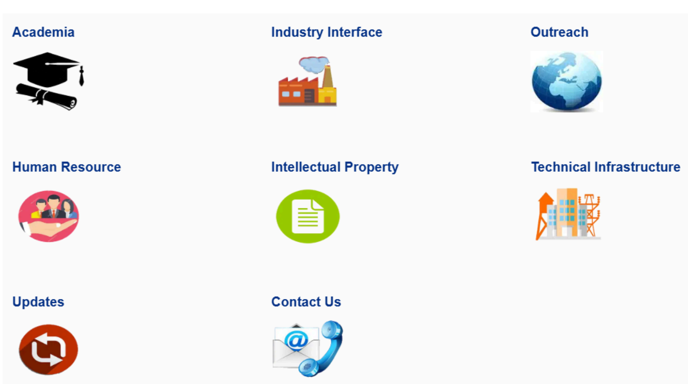


Fig. 2.2 no consistency in iconography

Recommendation: Adding those elements will increase user engagement and maintain consistency throughout the user journey

3.1.3 Layout, Content Hierarchy & Consistency

No grid system or alignment was followed, leading to a cluttered and inconsistent layout. Elements appear misaligned, reducing visual hierarchy (The Interaction Design Foundation, n.d.), and no consistency in shapes

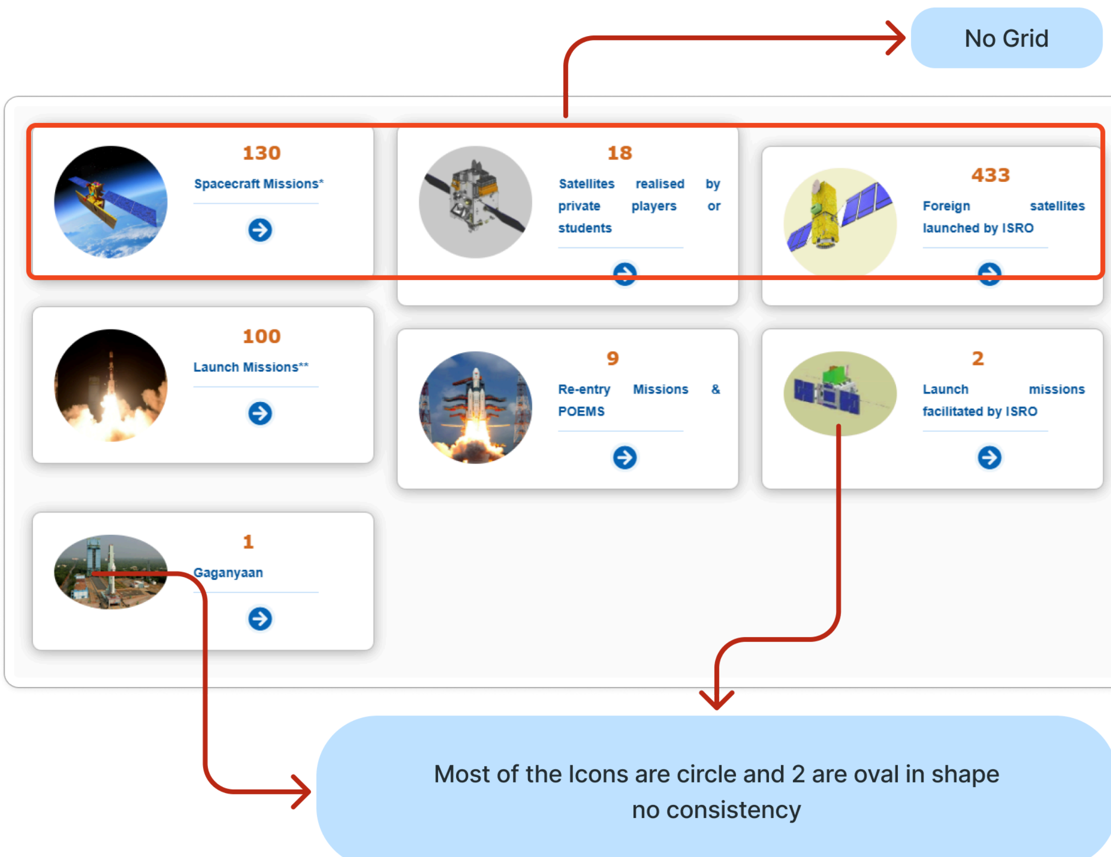


Fig. 2.3 Missions page with Menu

This inconsistency can **draw unintended attention** to that link, making users subconsciously perceive it as more important or different from the rest.

From a **UX perspective**, this could create confusion or disrupt visual harmony. If intentional, the design should reinforce its significance. According to the Von Restorff Effect, when multiple similar items are presented, the one that is different or stands out is more likely to be remembered. (Stevens, 2024)

User Heuristics: Flexibility and Efficiency of Use & Consistency and Standards (Nielsen, 2024)

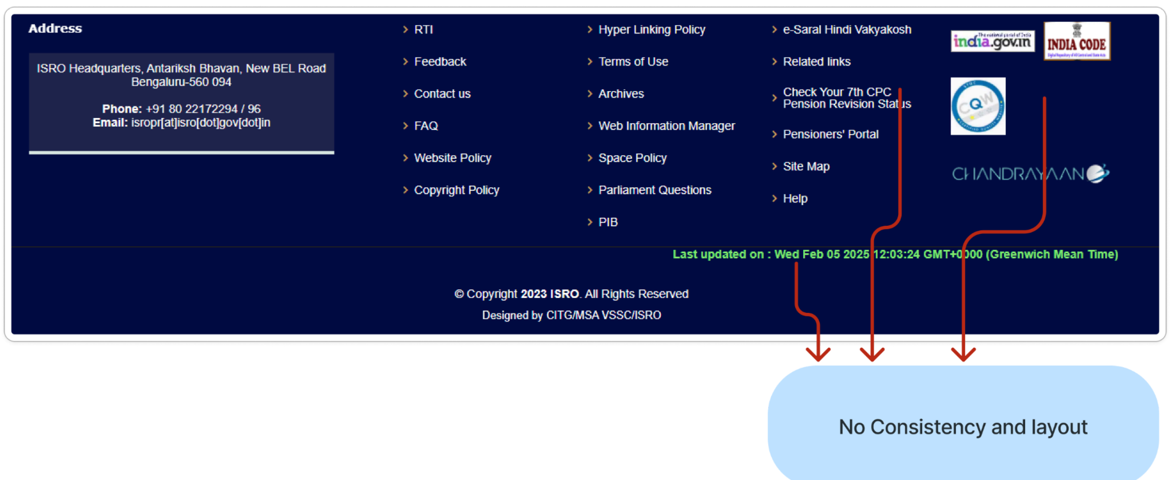


Fig. 2.4 footer with option and detail

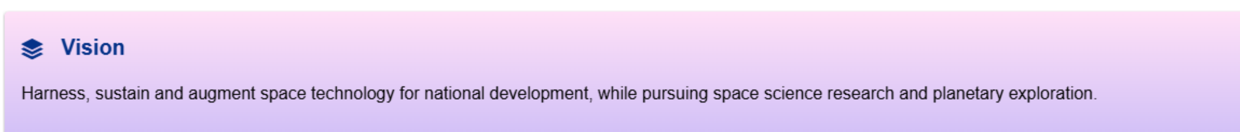


Fig. 2.5 One of the text body with a gradient background

And use gradients in many use cases are not consistent in showcasing the brand language

User Heuristics: Readability and Legibility (Nielsen, 2024)

Recommendation: Use 12 grid columns for alignment and spacing consistency and have a clear typographic hierarchy H1, H2 and standard spacing as uniform button shaps and spacing For the branding defined design system like colour palette typography and button styles and icon sets with similarity and avoid gradient and follow the brand colours for maximum accessibility

3.1.4 Readability & Body Content Consistency

The conventional navigation hierarchy is not intuitive, with text-heavy pages, excessive PDFs, and a non-sticky header that gets in the way of usability.

Compared to more user-centric websites like NASA and ESA, users may find it hard to locate critical information on time. Some clickable links are also hidden within the text, further making it hard to navigate.

User Heuristics: Readability and Legibility, User Control and Freedom (Nielsen, 2024)

According to Jakob's Law, Users spend most of their time on other websites, so they expect yours to work the same way. (Yablonski, 2020)

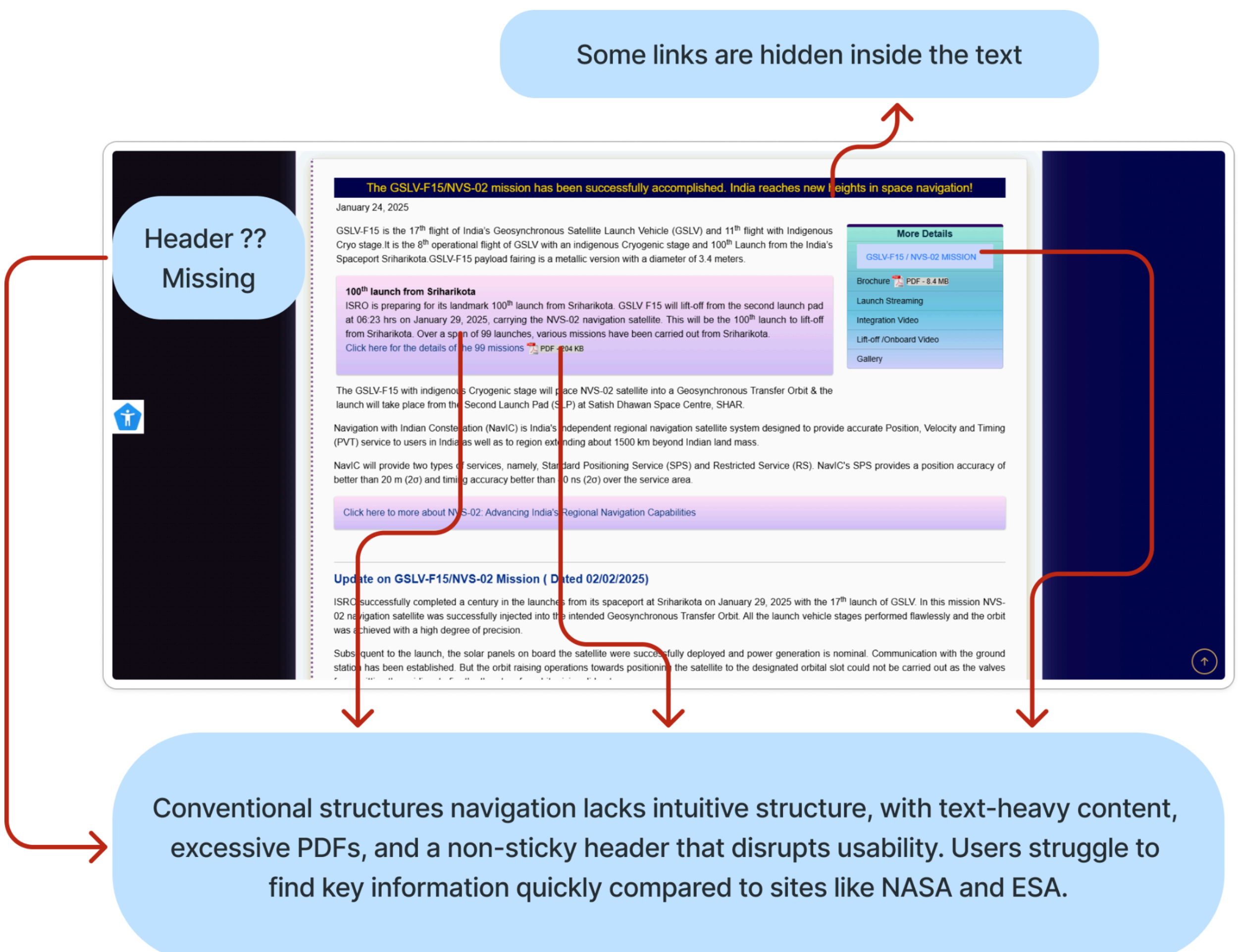


Fig. 2.6 Main body content with text-heavy and header missing

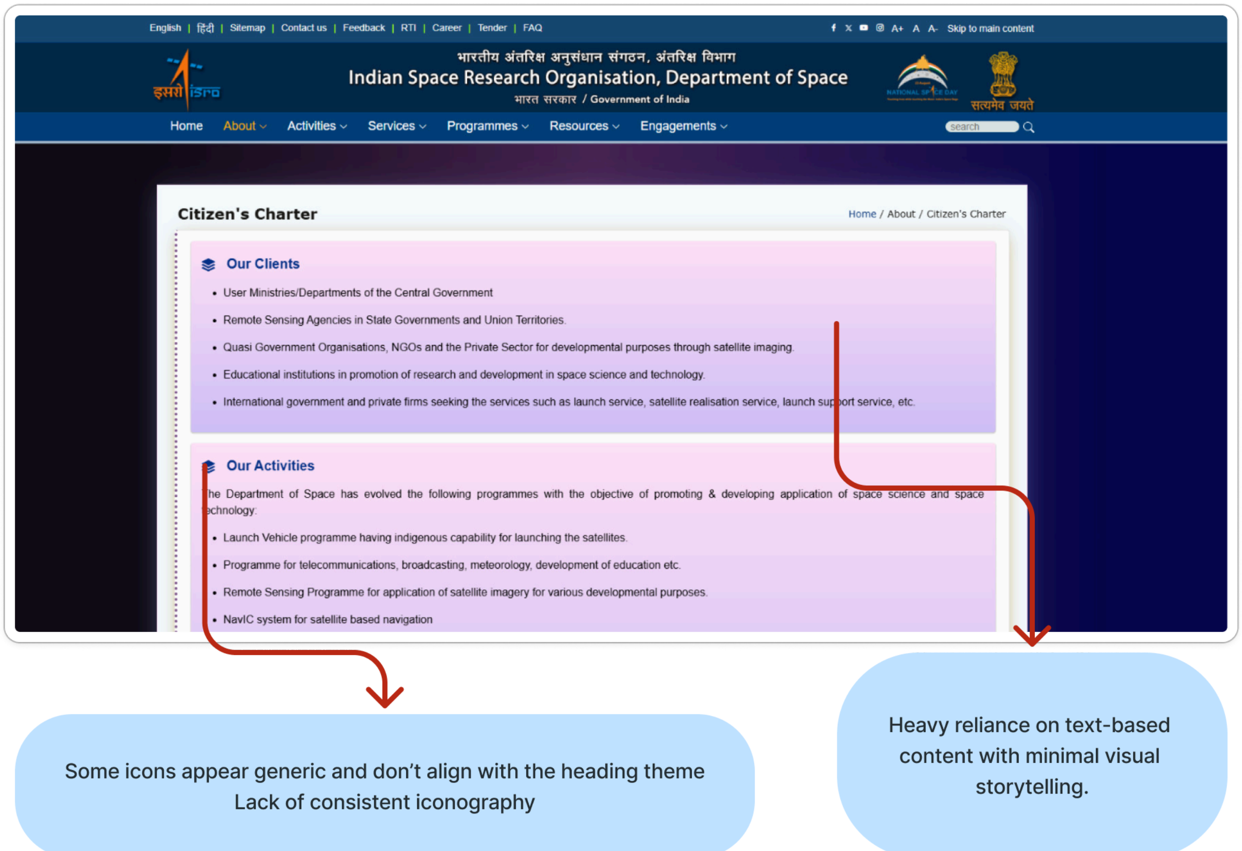


Fig. 2.7 Body content with no visual storytelling

Use the same kind of icon for every option and no visual storytelling to keep the engagement of the audience

Recommendation: To stick to user expectations, simplifying the layout complexity, improving navigation, and surfacing key links would make a big difference in improving usability and user experience and adding visual storytelling elements to improve user experience and user engagement

3.2 Accessibility Audit (Desktop & Mobile)

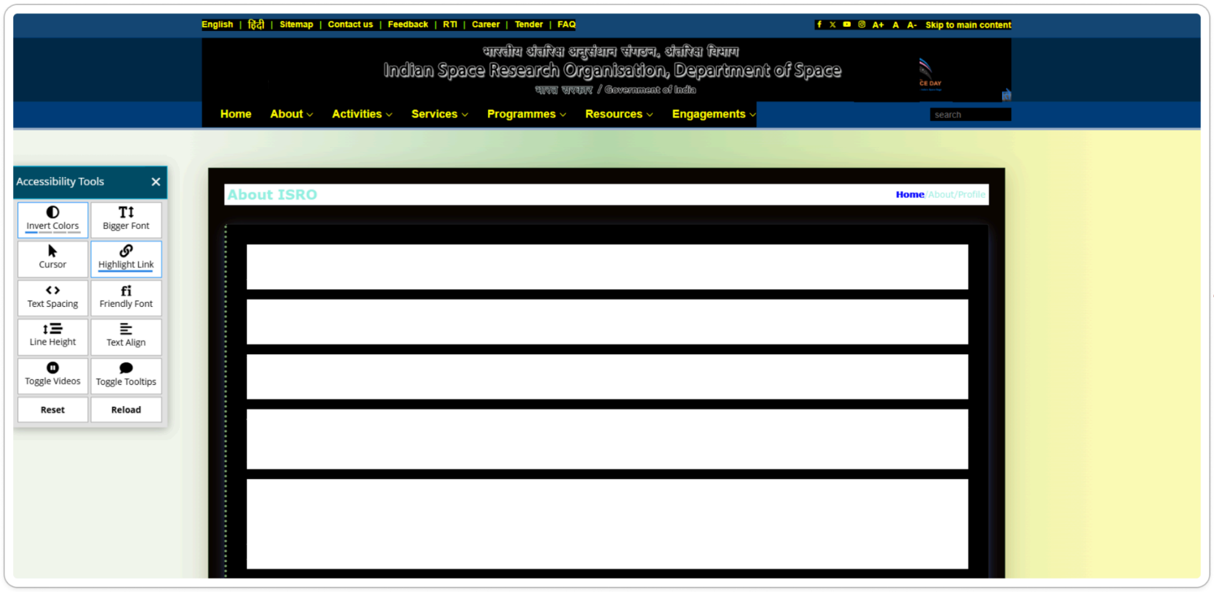
The accessibility option and the hardship encountered

3.2.1 Color Contrast & Typography (Legibility, WCAG compliance)

Color contrast Some text elements have insufficient contrast against background colors, making it difficult for users with visual impairments to read. Utilizing the accessibility tools will make the text invisible but not improve the visibility

Information and user interface components must be presented in ways that users can perceive (W3C, 2023)

User Heuristics: Match between system and the real world, Accessibility (Nielsen, 2024)



When accessibility tools invert colors and highlight links, the text disappears, making the website unusable.

Fig. 2.8 Accessibility tools block the text content

Recommendation: To have good contrast to the background colour and the text colour aligns with the brand identity and have the hierarchy and consistency

Certain sections failed in the colour contrast WCAG guidelines (W3C, 2023)

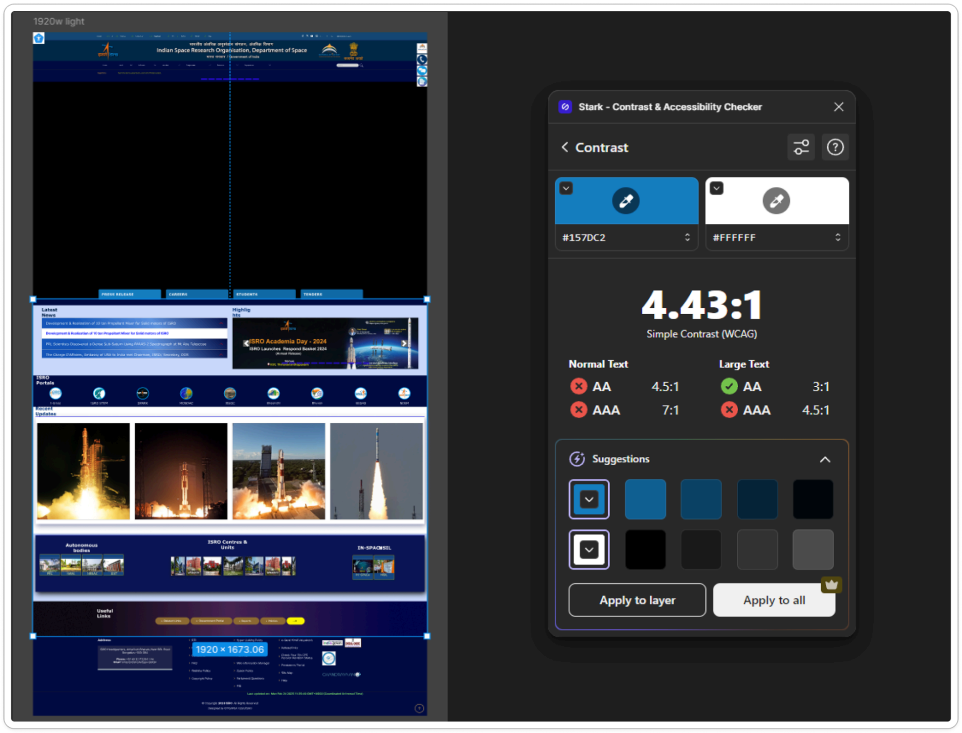
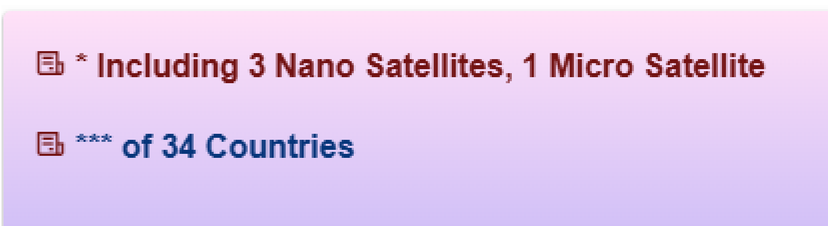


Fig. 2.9 Strack color contrast checking failed WCAG 2.2

The colour contrast for the slider menu for the flash news does not pass the WCAG guidelines



There is an inconsistency in the text colour, and it's a link to another page

Fig. 3.1 Text link within the background

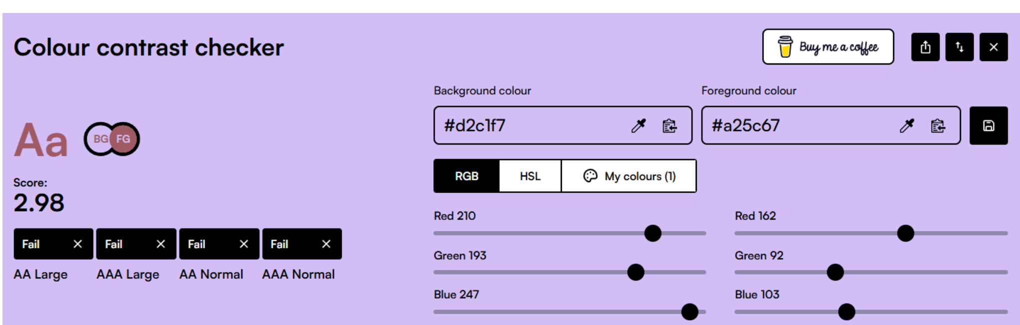


Fig. 3.2 Color contrast checker failed for the above text

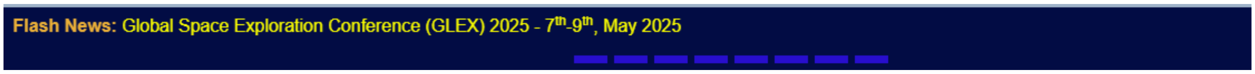


Fig. 3.3 Flash news slider

Following the flash news slider and menu the stopes contrast is barely visible using a gradient in the colour

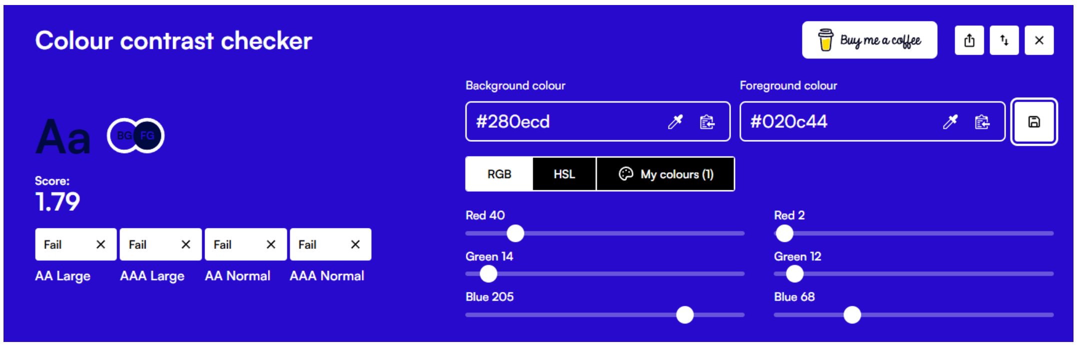


Fig. 3.4 Color contrast checker failed for the above text

3.2.2 Visuals & Multimedia Accessibility (Alt text, captions, video transcripts)

Multimedia In the case of videos, even if the captions are turned on, there is no caption in the video, and no transcripts are present. The website relies heavily on small and dense text, making scanning difficult.

Fig. 3.5 Video with no caption even the caption setting is ON

User Heuristics: Help users recognize, diagnose, and recover from errors, Flexibility and Efficiency of Use, Accessibility (Nielsen, 2024)

Recommendation: Adding a caption relevant with good contrast to the video will make the video accessible

3.2.3 Interactive Elements & Usability

Interactive controls and navigation must be operable by users (W3C, 2023)
This selection menu frustrates the user as the user can't move it to the desired option and needs to wait until it opens up to the option and it keeps rotating

User Heuristics: Error Prevention, User Control and Freedom (Nielsen, 2024)

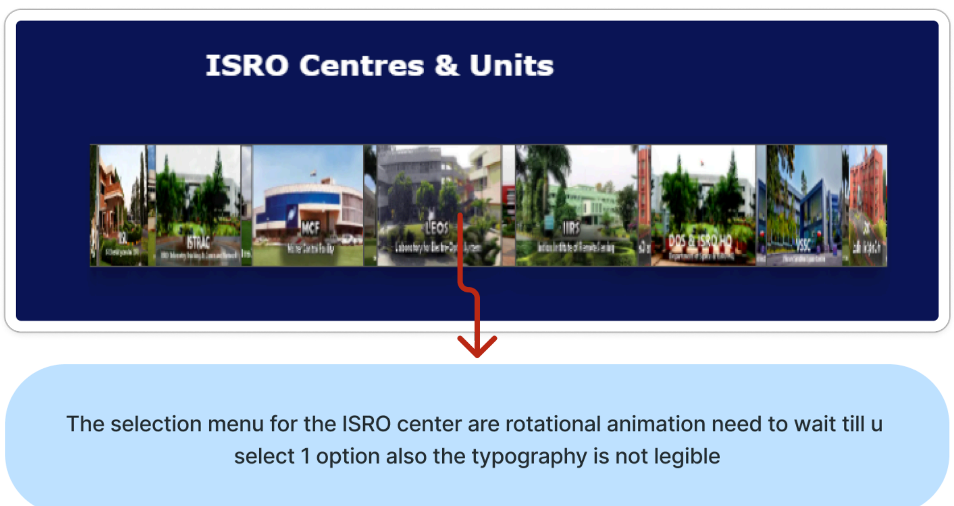


Fig. 3.6 Unusable rotating menu

Recommendation: To create a clear and easy-to-use menu

3.2.4 Keyboard Navigation & Assistive Tech Support

Fig. 3.7 Keyboard navigation doesn't work in certain sections like the header

Some of the options for keyboard navigation work, and some will just skip the whole content or select a bunch of content at the same time, so it is not operable by the keyboard

User Heuristics: Consistency and Standards, Accessibility (Nielsen, 2024)

Recommendation: Add alt text and add keyboard navigation for the sessions which are missing

3.2.5 Content Structure

Miller's Law states that the average person can hold about 7 (±2) items in their working memory at a time. (Yablonski, 2020) ISRO's website, being text-heavy with hardly any interactive elements, is difficult for users who are trying to absorb information. With such a lot of text, users will find it hard to retain and process key information easily, thereby causing cognitive overload.

User Heuristics: Recognition rather than Recall, Flexibility and Efficiency of Use (Nielsen, 2024)

To improve user experience, simplifying content, breaking up text into small chunks, and adding more interactive features would enable users to focus on the most important information without overwhelming their memory. This would make the website more accessible and make it easier to digest content.

Accessibility & Readability **Law of Prägnanz** Simplicity in Design Use clear and concise visual elements like simple shapes, minimal text and intuitive icons. **(The Interaction Design Foundation, n.d.)** and no Headline in most of the text content

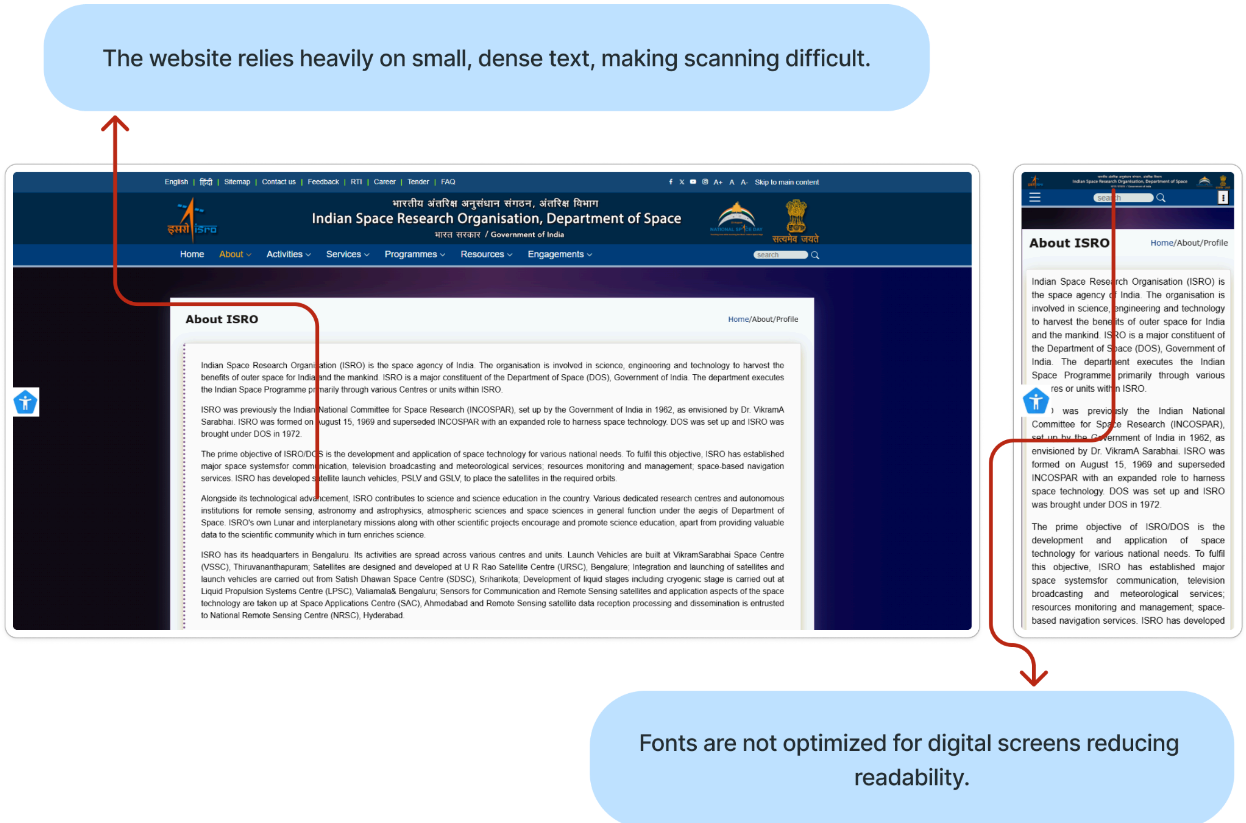


Fig. 3.8 Text-heavy content with no visual storytelling and huge chunks of the entire page

Recommendation: reduce the text and create it into smaller chunks to make the use clear and introduce hierarchy for the text

Representation of Women & Minorities - The website sometimes features content about women in STEM, but broader representation can be improved.
Accessibility for Differently-Abled Users - No dedicated section for accessibility features or assistive tech guidelines. But has an Accessibility toolbar but it's not well developed

3.2.6 Mobile and Desktop Version

The mobile version is not optimized the layout feels cluttered the text and image are not well-scaled and there is no dedicated app for mobile and acks responsiveness

User Heuristics: Responsive Design & Flexibility (Flexibility and Efficiency of Use) Consistency Across Devices (Consistency and Standards) (Nielsen, 2024)

3.3 Usability and Performance

3.3.1 Page Load Speed & Optimization

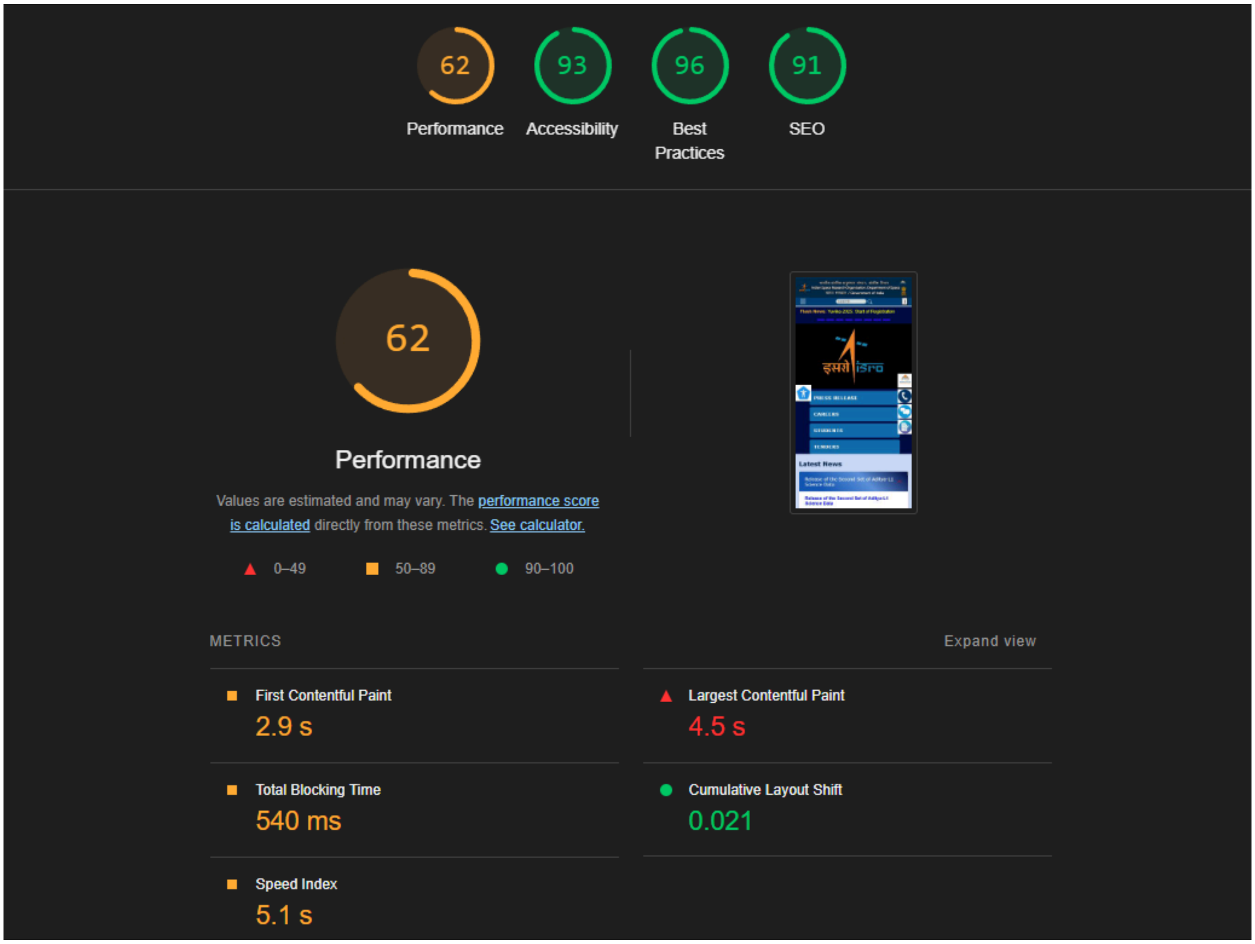


Fig. 3.9 Google lighthouse results for page loading speed and accessibility

Loading Speed: The website experiences slow loading times, particularly on pages with heavy media content. Like photos and videos Usually, it is faster when it comes to the mobile website having huge text-heavy pages and desktop speed is good

Interactive Elements: The site provides informative resources but lacks dynamic engagement features such as interactive timelines or data visualisation items.

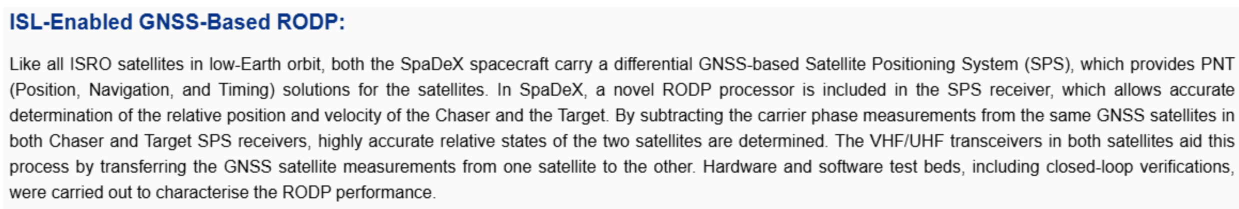
Search Functionality: The search bar works but is not highly intuitive, and some queries return irrelevant or outdated results. And doesn't show FAQ and has no suggestions

3.3.2 Language Clarity and Multilingual Accessibility

Multiple Language Support – ISRO provides Hindi and English, focused on a large audience. Given India's linguistic diversity, adding more regional languages could improve inclusivity.

Clear language: Simple, clear, and concise language to communicate the content. The ISRO website uses lots of jargon in text content, especially in complicated space terminology.

Understandable: The user interface information and operation should be understandable. (W3C, 2023)



ISL-Enabled GNSS-Based RODP:

Like all ISRO satellites in low-Earth orbit, both the SpaDeX spacecraft carry a differential GNSS-based Satellite Positioning System (SPS), which provides PNT (Position, Navigation, and Timing) solutions for the satellites. In SpaDeX, a novel RODP processor is included in the SPS receiver, which allows accurate determination of the relative position and velocity of the Chaser and the Target. By subtracting the carrier phase measurements from the same GNSS satellites in both Chaser and Target SPS receivers, highly accurate relative states of the two satellites are determined. The VHF/UHF transceivers in both satellites aid this process by transferring the GNSS satellite measurements from one satellite to the other. Hardware and software test beds, including closed-loop verifications, were carried out to characterise the RODP performance.

Fig. 4.1 Jargon text is thought out on the website

Recommendation: Remove the text jargon and make it simple and understandable to everyone

3.3.3 Social Media and Connect

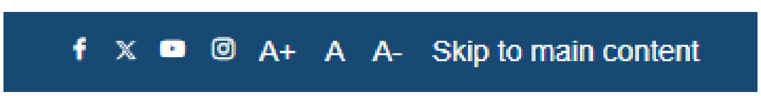


Fig. 3.8 Only option to access the official social account

ISRO doesn't have any option to follow their socials except the top header had social with text size increase option which is

The pixel size is smaller than the WCAG guidelines for text, which is 12px. There is no showcase or promotion of the official social media accounts

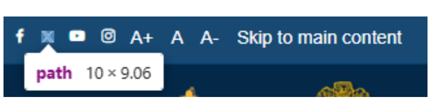


Fig 4.3 Only option to access the official social account

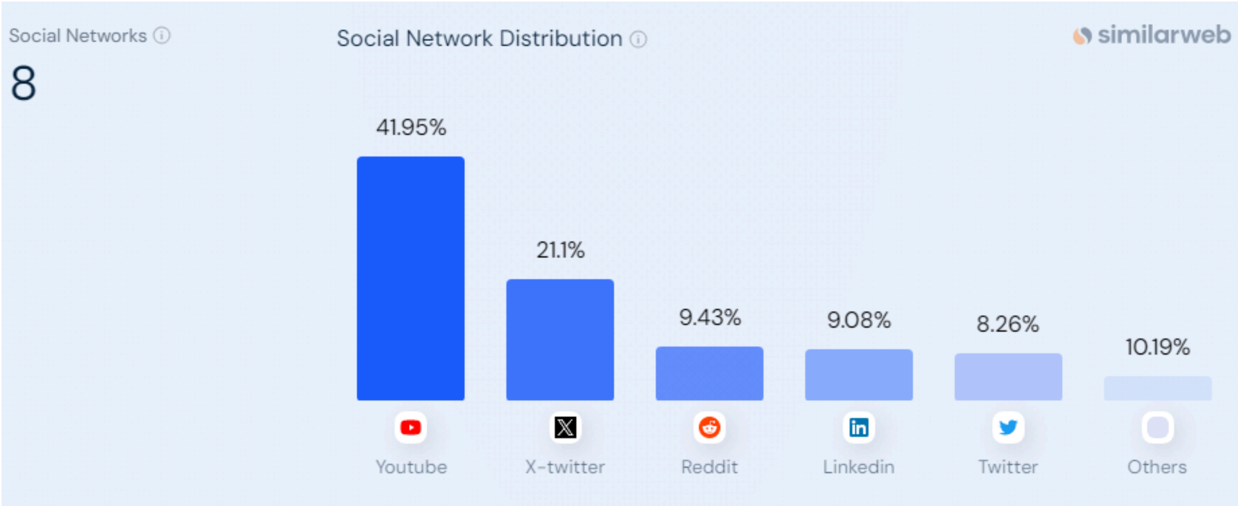


Fig. 4.4 Social network distribution

Recommendation: create a section which promotes social media and shows updates it's easy to consume info through social media

4. Comparative Analysis

To understand how ISRO's website performs in terms of usability, design, accessibility, and functionality, we compared it with:

NASA - https://www.nasa.gov/ - U.S. Space Agency (NASA, 2025)

ESA - https://www.esa.int/ - European Space Agency (European Space Agency, 2019)

Roscosmos - https://www.roscosmos.ru/ - Russian Space Agency (www.roscosmos.ru, n.d.)

Criteria	ISRO (India)	NASA (USA)	ESA (Europe)	Roscosmos (Russia)
Visual Design & Branding	Simple, outdated UI; lacks interactive elements	Modern, clean, visually appealing	Professional, well-structured, high-quality images	Overloaded layout, inconsistent branding
Navigation & UX	Text-heavy, difficult to browse quickly	User-friendly, structured categories	Intuitive menu, strong organisation	Complex navigation lacks English support
Accessibility & Responsiveness	Some accessibility issues, lack of mobile optimization	WCAG-compliant, mobile-friendly	Responsive, colour-contrast optimised	Not fully mobile-friendly
Content & Engagement	Informative but lacks interactive storytelling	Interactive mission pages, rich multimedia	A mix of articles, videos, and infographics	Heavy text-based content
Ordering & Checkout Flow	Limited / No e-commerce functionality	Seamless NASA store with filtering & checkout	Well-organized ESA shop, smooth checkout	No public-facing merchandise shop
Search & Discoverability	Basic search lacks filters	Advanced search & AI-powered recommendations	Categorized archives, easy filtering	Search is slow and inefficient
Rating	1.5/5	4.5/5	3.5/5	3/5

Fig. 4.5 Comparative Analysis Chart

NASA sets the **gold standard** with interactive elements, clear navigation, and a strong brand identity.

ISRO can enhance its UI with engaging infographics, mission timelines, and multimedia integration.

ESA's organization and accessibility are superior to ISRO

ISRO should adopt structured content layouts, colour contrast improvements, and mobile-first design to improve usability.

Roscosmos struggles with navigation, but its detailed technical content is useful.

ISRO can benefit from categorizing scientific resources for better public and academic access.

5. Conclusion

In conclusion, the ISRO website delivers information but requires significant enhancements in usability, accessibility, and modern UX/UI design.

Improving navigation The ISRO website is an excellent resource for space knowledge but falls behind on usability and aesthetics. Making accessibility, and navigation to make it more user-friendly. While the content and text are comprehensive, especially across devices. Streamlining the interface would enable users to find the information they want swiftly.

Adopting modern designs like minimalist structures, cleaner fonts, and more vibrant visuals would improve aesthetics as well as usability. Further, simplifying accessibility features like text-to-speech options, contrast modes, and keyboard navigation would open the site to a wide range of audiences. To get the website to up-to-date UI/UX And features

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7. List of Figures

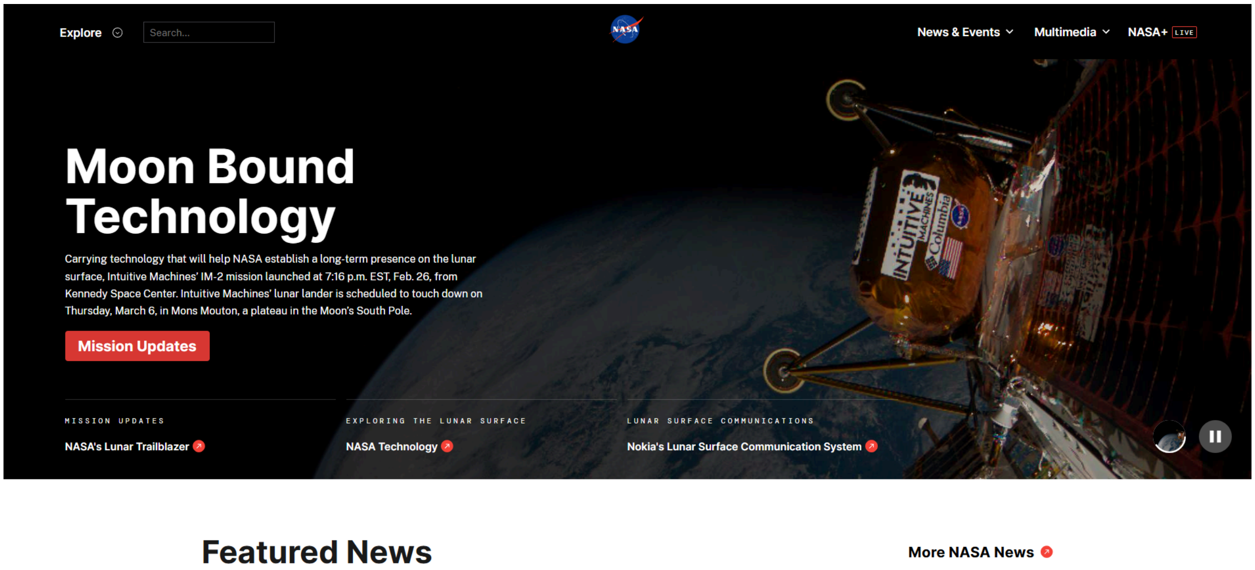
8. Appendix

Comparative study

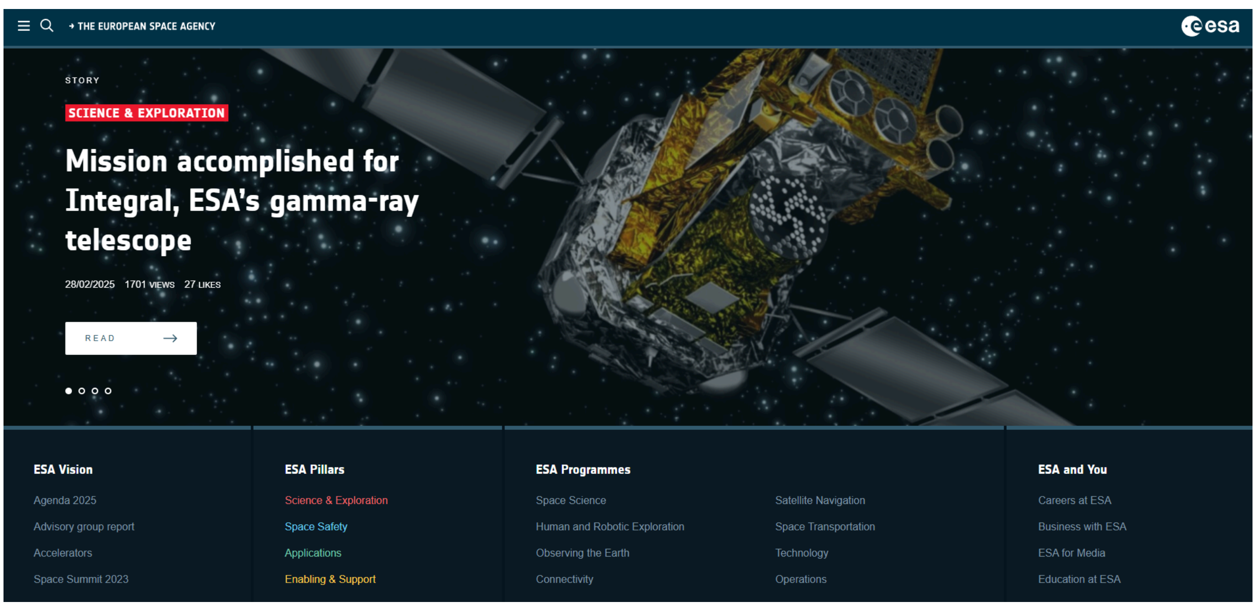
The first impression on the websites

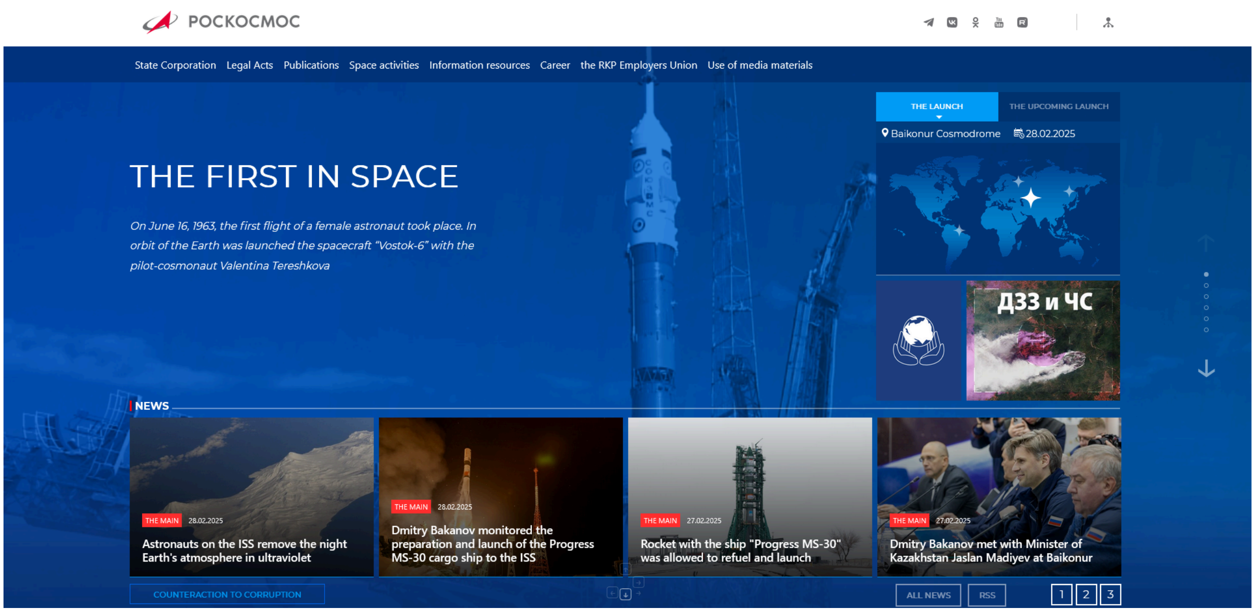
NASA - https://www.nasa.gov/ - U.S. Space Agency (NASA, 2025)

Roscosmos - https://www.roscosmos.ru/ - Russian Space Agency (www.roscosmos.ru, n.d.)



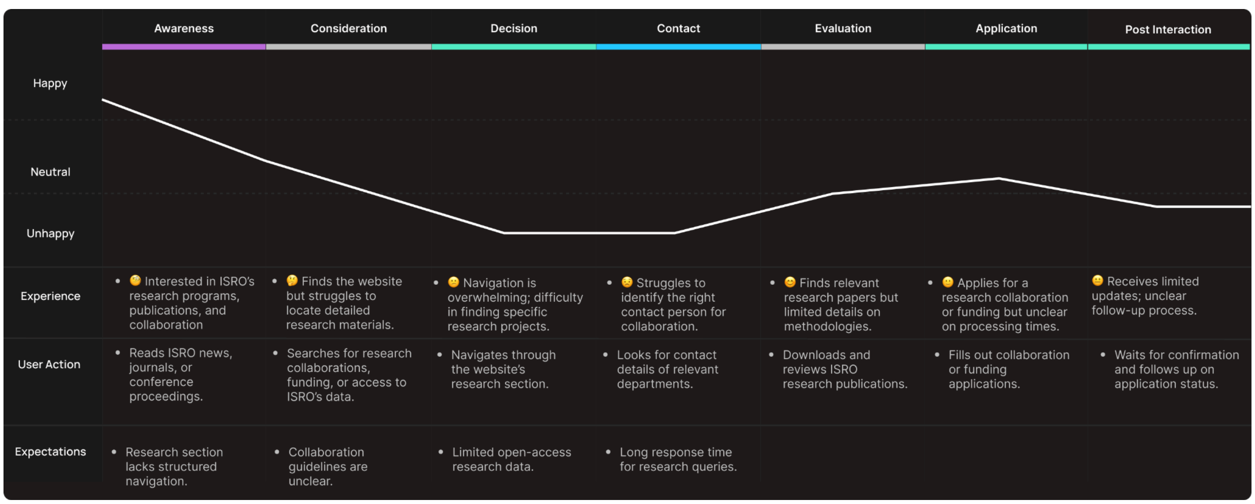
ESA - https://www.esa.int/ - European Space Agency (European Space Agency, 2019)



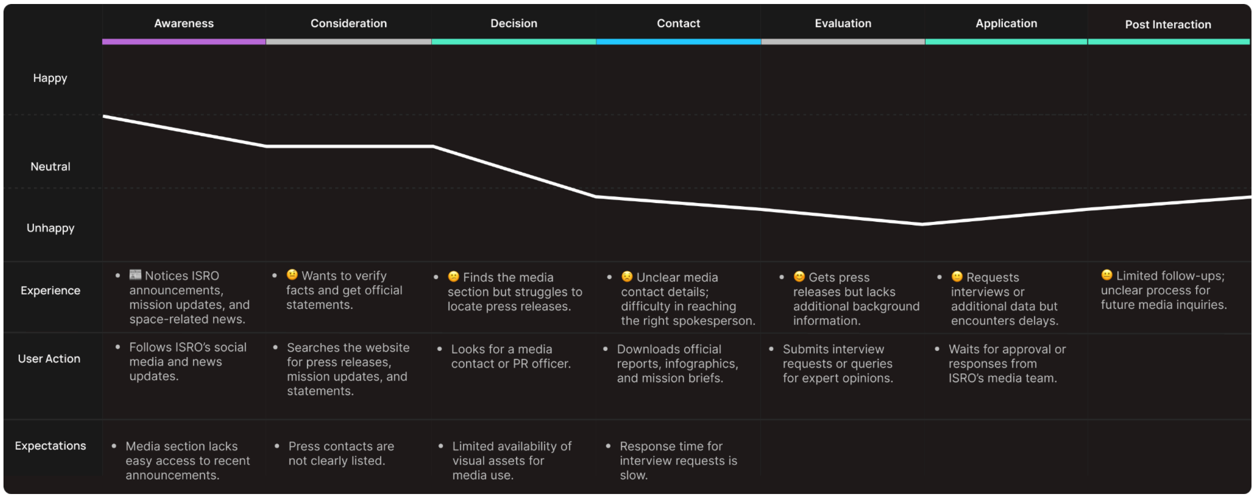


Roscosmos - https://www.roscosmos.ru/ - Russian Space Agency (www.roscosmos.ru, n.d.)

User Journey map (Nielsen Norman Group, 2019)



User journey map of a researcher exploring opportunities and collab



User journey map of a journalist exploring opportunities and collab